

**KINGSCLIFF
HIGH SCHOOL**



ACHIEVEMENT THROUGH ENDEAVOUR

STAGE 4 · YEAR 7 MATHEMATICS

Data Analysis

Lesson 1 · Mean, median and mode

Name

Class

Teacher

How to use this booklet

Each lesson follows the same steps. Work through them in order and write everything in this booklet.

Teacher example

Your teacher works through the question on the board. Copy the working into the box.

Your turn

Answer the questions yourself. Show your working in the box and write your final answer on the line.



The dots show the level. One dot is getting started, two is building up and three is a challenge.



Extension questions stretch your thinking further. Read the worked example first.

Summary

Missed a lesson? The summary pages have a fully worked example for every level.

Exit ticket

Choose one question to show off your skill, then tear it off and hand it to your teacher.

What we are learning

NSW MATHEMATICS K-10 SYLLABUS (2022) · STAGE 4 · STATISTICS · DATA ANALYSIS

MA4-DAT-C-02: analyses simple datasets using measures of centre, range and shape of the data

CONTENT FOCUS FOR THIS LESSON

- Calculate the mean, median and mode for sets of data and interpret them in the context of the data
- Investigate the effect of individual data values, including outliers, on the mean and median



Acknowledgement of artwork

The artwork used throughout this booklet was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Learning intentions

Today we are learning to:

- understand what a **measure of centre** is and why we use one
- calculate the **mean**, **median** and **mode** of a set of data
- use these measures to solve problems about real data

Success criteria

I can:

- ☐ find the **mean** by adding the values and dividing by how many there are
- ☐ find the **median** by ordering the data and finding the middle value
- ☐ find the **mode**, and tell when there is no mode or more than one
- ☐ work backwards from the mean to find a missing value

Key terms

Write the definition as your teacher explains it.

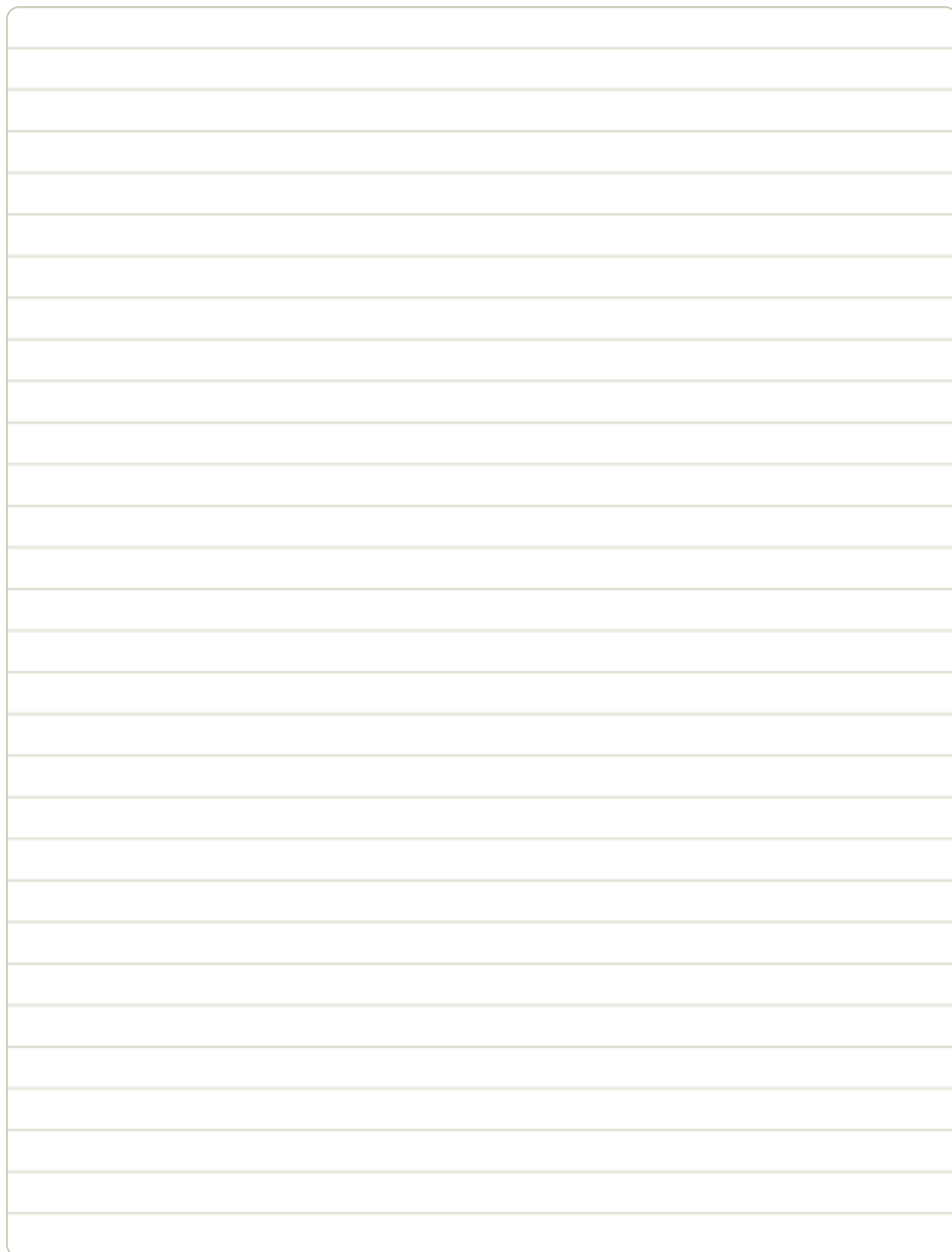
Mean

Median

Mode

My notes

Write notes, diagrams and reminders from the lesson here.



Teacher example: copy the working as your teacher shows you

Example 1 Find the **mean** of these numbers: 4, 7, 5, 8, 6

Example 2 Find the **median** and the **mode** of these numbers: 3, 9, 4, 7, 4, 8, 5

Level 1 · Getting started

Your turn: Set A. Find the mean of each set of numbers.

1 2, 4, 6

Answer: _____

2 5, 7, 9

Answer: _____

3 1, 3, 5, 7

Answer: _____

4 10, 20, 30

Answer: _____

5 3, 5, 4, 8

Answer: _____

6 6, 6, 9, 3

Answer: _____

7 2, 8, 5, 7, 3

Answer: _____

8 12, 15, 18

Answer: _____

9 4, 9, 2, 5, 10

Answer: _____

10 7, 11, 9, 13

Answer: _____

Level 1 · Getting started

Your turn: Set B. Find the median and the mode of each set. Order the numbers first!

1 2, 3, 3, 5, 8

Median: _____ Mode: _____

2 1, 4, 4, 6, 9

Median: _____ Mode: _____

3 7, 2, 5, 2, 9

Median: _____ Mode: _____

4 6, 1, 8, 6, 3

Median: _____ Mode: _____

5 10, 12, 11, 10, 15

Median: _____ Mode: _____

6 4, 9, 7, 9, 2, 5, 9

Median: _____ Mode: _____

7 15, 13, 14, 13, 18

Median: _____ Mode: _____

8 8, 3, 6, 3, 1, 5, 3

Median: _____ Mode: _____

9 20, 25, 22, 25, 21

Median: _____ Mode: _____

10 5, 7, 6, 7, 9, 4, 8

Median: _____ Mode: _____

Teacher example: copy the working as your teacher shows you

Example 3 Find the **median** of: 8, 3, 10, 5, 7, 12

Example 4 A basketball player scored 12, 15, 9, 18, 11 and 13 points in six games. Find the **mean**, **median** and **mode** of her scores.

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

**Level 2 · Building up****Your turn: Set C. Show your working in each box.**

- 1** Find the **median** of: 4, 9, 2, 7

Answer: _____

- 2** Find the **median** of: 11, 15, 12, 18, 14, 20

Answer: _____

- 3** Find the **mean** of: 3, 8, 5, 6

Answer: _____

- 4** Find the **mean** of: 2.5, 3.5, 4, 6

Answer: _____

Your turn: Set C continued.

- 5 Find the **mode** of: 6, 2, 8, 2, 6, 5

Answer: _____

- 6 The heights of six students (in cm) are 150, 145, 160, 155, 148 and 152. Find the **median** height.

Answer: _____

- 7 The maximum temperatures ($^{\circ}\text{C}$) for a week were 21, 24, 19, 24, 22, 23, 21. Find the **mean**, **median** and **mode**.

Answer: _____

Your turn: Set C continued.

- 8 A soccer team scored 2, 0, 3, 1, 2, 4, 2, 2 goals in eight games. Find the **mean, median and mode**.

Answer: _____

- 9 Find the **mean** of: 7, 9, 10, 12, 14

Answer: _____

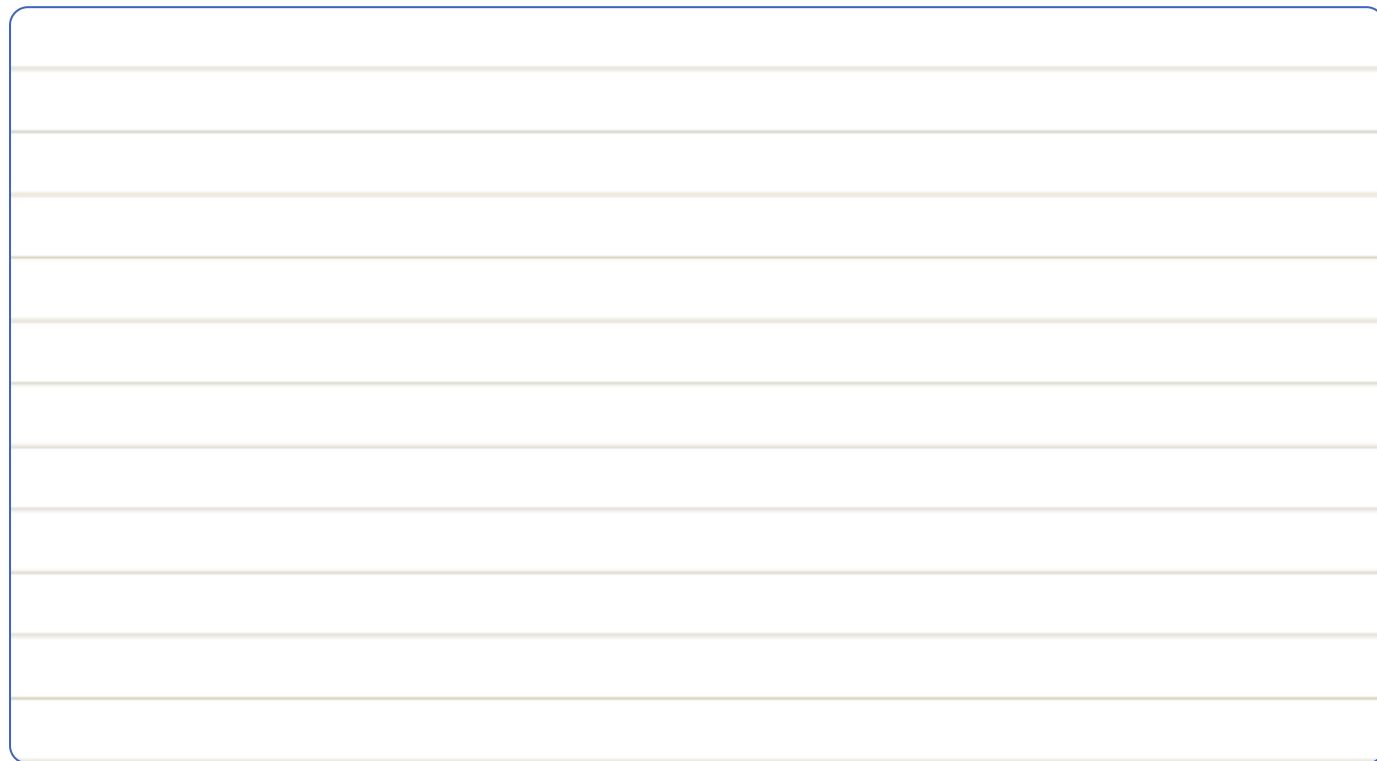
- 10 Six friends get \$15, \$20, \$10, \$25, \$20 and \$30 pocket money each week. Find the **mean, median and mode**.

Answer: _____

 Level 3 · Challenge

Teacher example: copy the working as your teacher shows you

Example 5 The mean of five numbers is 8. Four of the numbers are 6, 9, 5 and 10. Find the **fifth number**.



Example 6 The ages of six people at a party are 11, 12, 12, 13, 12 and 44. Find the **mean** and **median** age. Which one better describes a typical age at the party? Explain why.



 Level 3 · Challenge**Your turn: Set D. Show your working in each box.**

- 1 The mean of four numbers is 6. Three of the numbers are 4, 7 and 8. Find the fourth number.

Answer: _____

- 2 The mean of five test scores is 72. Four of the scores are 68, 75, 70 and 80. Find the fifth score.

Answer: _____

- 3 Write a set of five numbers that has a mean of 6, a median of 5 and a mode of 4.

Answer: _____

Your turn: Set D continued.

- 4 For the data 3, 5, 6, 6, 40, find the mean and the median. Which is the better measure of centre? Explain why.

Answer: _____

- 5 The number 20 is added to the data set 4, 6, 7, 8, 10. How do the mean and the median change?

Answer: _____

- 6 Mia scored 7, 8, 6 and 9 on four quizzes. What does she need to score on her fifth quiz to have a mean of 8?

Answer: _____

Your turn: Set D continued.

- 7 Three whole numbers have a mean of 10 and a median of 9. The smallest number is 5. Find the largest number.

Answer: _____

- 8 The mean of six numbers is 15. When one number is removed, the mean of the other five is 14. Which number was removed?

Answer: _____

- 9 A shoe shop sold these sizes today: 7, 8, 8, 9, 10, 8, 11. The owner wants to know which size to order the most of. Should she use the mean, median or mode? Explain.

Answer: _____

- 10 Class A scored 12, 15, 15, 18, 20 and Class B scored 10, 14, 16, 19, 21 on a quiz. Compare the two classes using the mean and the median.

Answer: _____



Extension · Stretch yourself

Read the worked example carefully, then try the extension questions.



Worked example

The mean age of four friends is 13. When a fifth friend joins the group, the mean age becomes 14. How old is the fifth friend?

1. Find the total of the four ages: $4 \times 13 = 52$
2. Find the total of all five ages: $5 \times 14 = 70$
3. The difference is the new friend's age: $70 - 52 = 18$
4. Check: $(52 + 18) \div 5 = 70 \div 5 = 14 \checkmark$

Answer: the fifth friend is 18 years old.

Key idea: mean $\times$ number of values = total. Working with totals makes “missing value” questions much easier.

- E1** The mean mark of 10 students on a test is 65. The teacher re-marks one test and adds 15 marks to it. What is the new mean?

Answer:

Your turn: Extension continued.

- E2** A list of five positive whole numbers has a mean of 7, a median of 8 and only one mode, which is 9. Find **all** the possible lists.

Answer: _____

- E3** The mean of three consecutive whole numbers is 17. (a) What are the numbers? (b) Explain why the mean of any three consecutive whole numbers is always the middle number.

Answer: _____

Lesson summary

Missed the lesson? Start here. These pages show you everything you need, step by step.

Mean

The **average**. Add up all the values, then divide by how many values there are.

$$\text{mean} = \frac{\text{total of the values}}{\text{number of values}}$$

Median

The **middle value** when the data is written in order from smallest to largest.

Two middle values? The median is halfway between them.

Mode

The value that appears **most often**.

There can be no mode, or more than one mode.

Finding the median: cross off from both ends

Put the data in order, then cross off one value from each end until you reach the middle.

Odd number of values
one middle value



median = 9

Even number of values
two middle values



median = $(8 + 10) \div 2 = 9$

Which measure should I use?

Mean

Uses every value. Best when there are no unusually large or small values.

e.g. quiz scores 6, 7, 8, 9: mean = 7.5

Median

Not pulled around by an **outlier** (a value much bigger or smaller than the rest). Best when the data has an outlier.

e.g. ages 11, 12, 12, 13, 44: median = 12, but the mean is 18.4

Mode

Shows the most common value. Best for finding the most popular choice, like a shoe size or favourite colour.

e.g. shoe sizes sold 7, 8, 8, 9, 8: mode = size 8

 Level 1 worked example

Find the mean, median and mode of: 3, 5, 7, 5, 10

1. **Mean:** add up the values: $3 + 5 + 7 + 5 + 10 = 30$
2. Count the values: there are 5 values
3. Divide the total by the count: $30 \div 5 = 6$
4. **Median:** write the values in order: 3, 5, 5, 7, 10
5. Cross off from both ends. The middle (3rd) value is 5
6. **Mode:** 5 appears twice. Every other value appears once, so the mode is 5

Answer: mean = 6, median = 5, mode = 5

 Level 2 worked example

Find the mean, median and mode of: 9, 4, 12, 6, 10, 7

1. **Mean:** add up the values: $9 + 4 + 12 + 6 + 10 + 7 = 48$
2. There are 6 values, so the mean is $48 \div 6 = 8$
3. **Median:** write the values in order: 4, 6, 7, 9, 10, 12
4. There are 6 values (an even number), so there are **two** middle values: 7 and 9
5. Find halfway between them: $(7 + 9) \div 2 = 16 \div 2 = 8$
6. **Mode:** every value appears only once, so there is **no mode**

Answer: mean = 8, median = 8, no mode

 Level 3 worked example

The mean of four test scores is 15. Three of the scores are 12, 18 and 14. Find the fourth score.

1. **Total of all four scores:** $\text{mean} \times \text{number of scores} = 15 \times 4 = 60$
2. **Total of the three known scores:** $12 + 18 + 14 = 44$
3. **Missing score:** $60 - 44 = 16$
4. **Check:** $(12 + 18 + 14 + 16) \div 4 = 60 \div 4 = 15 \checkmark$

Answer: the fourth score is 16

 Extension worked example

The mean of five numbers is 10. A sixth number is added and the mean becomes 12. What number was added?

1. **Total of the five numbers:** $5 \times 10 = 50$
2. **Total of the six numbers:** $6 \times 12 = 72$
3. **Number added:** $72 - 50 = 22$
4. **Check:** $72 \div 6 = 12 \checkmark$

Answer: the number added was 22

Common mistakes to avoid

- Forgetting to **put the data in order** before finding the median.
- Dividing by the wrong number when finding the mean. **Count the values carefully.**
- Giving the **position** of the median (e.g. “the 3rd value”) instead of the value itself.
- Thinking there must always be one mode. There can be **none**, or **more than one**.
- An **outlier** pulls the mean towards it. The median is not affected as much.

Tick one box for each success criterion.

I can...	Not yet	Nearly	Got it!
find the mean of a data set	☐	☐	☐
find the median, including when there are two middle values	☐	☐	☐
find the mode, and tell when there is no mode or more than one	☐	☐	☐
work backwards from the mean to find a missing value	☐	☐	☐

----- ✂ Tear or cut along this line and hand the bottom part to your teacher ✂ -----

Name	Class
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education
John	Language Arts
Jane	Language Arts
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education
John	Language Arts
Jane	Language Arts

Level 1	Level 2	Level 3
Find the mean of: 4, 6, 8, 10, 12	Find the median and mode of: 9, 3, 7, 3, 11, 6	The mean of four numbers is 9. Three of them are 7, 12 and 5. Find the fourth number.
Answer: _____	Answer: _____	Answer: _____

Questions I still have

Write anything you are still unsure about. Ask your teacher about it next lesson.

----- ✂ -----

This space is left blank on purpose.
It is the back of your exit ticket.